Galaba Vamsi

Software Engineer (Full-stack, AI Agents), Intern / New Grad (graduating May 2027)
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SUMMARY

Mechatronics at IIT Bhilai. I ship full-stack products and AI agents: co-founder of Human Slop (live on the web and Google Play) and Robotropic; Charter, an MCP commerce agent; a guarded LLM chatbot with a measured eval; top 10 of 4,500+ teams at RAKSHAM. 15 merged open-source PRs (11 code) and a maintained PyPI package.

EDUCATION

B.Tech. in Mechatronics, Indian Institute of Technology (IIT) Bhilai Aug 2023 - May 2027
Coursework: DSA, OOP, AI and ML, Deep Learning for Computer Vision, Digital Image Processing.

EXPERIENCE

Co-founder, Robotropic Physical-AI startup, with Varun Rao | robotropic.com Oct 2026 - Present
- Building Harness, a workspace where a factory's team and AI agents change robot skills safely: each change is checked against safety limits, rolled out cell by cell, and recorded. I lead the simulated factory floor and the change gateway.
- Built robotropic.com: three.js lidar factory scene, early-access list on Cloudflare Workers and D1.

Co-founder, Human Slop Anti-AI social app, with Varun Rao | humanslop.in Mar 2026 - Present
- Expo (React Native) app for web and Android, Fastify API, PostgreSQL on Supabase, deployed on Render and Cloudflare; live on the web and Google Play (50+ downloads). Built a signup biometric check and an editor keystroke-timing score (heuristics).

Research Intern, IIT Bhilai Advisor: Dr. Arzad Alam Kherani May 2025 - Jul 2025
- Built RIS-SIM, an open-source Python radio-link emulator with USRP-style SDR interfaces (later rebuilt as sremu); co-author, IEEE INDICON 2025.

PROJECTS

Charter (Razorpay AI Buildathon, 2026) Agentic commerce: chat, voice, and a 10-tool MCP server call one first-party API, so agents get no side door; only versioned policy changes money (frozen INR quotes, Razorpay test mode, PostgreSQL row-level security). TypeScript, Fastify, React; deployed to AWS Lightsail via GitHub Actions with OIDC. charter.humanslop.in (live, test mode) | github.com/Galabavamsi/charter

BIT, guarded portfolio chatbot (2026) LLM chatbot behind a Supabase Edge Function (TypeScript) with a typed decision model (TypeSafe Jev) as the gate. On 50 labelled messages it blocked 0 of 20 real questions (keyword rules: 12) and let 0 of 12 prompt-injection attempts through (regex: 10); median gate latency 304 ms. galabavamsi.github.io/portfolio/blog

Homie (2026) Collaborative food-ordering MVP on Swiggy's MCP server: Flutter (Riverpod) app, Node.js (Express) REST API, PostgreSQL, and Socket.IO realtime rooms over Redis for multi-instance delivery; voting, shared cart, bill split, idempotent operations; OAuth 2.1 + PKCE with tokens kept on the server. github.com/Galabavamsi/homie

Open-PyFX (2026) GPU video-effects app in Python, ModernGL, and GLSL shaders: CRT emulation, retro text, ASCII edge detection, and 4K export. github.com/Galabavamsi/open-pyfx

sremu (2026, MIT, built with Claude Code) Python IQ-level radio emulator with a ZeroMQ data plane that unmodified GNU Radio 3.10 talks to; sample-exact timestamped commands; about 2,140 tests; benchmarked on AWS; sole-author preprint draft. github.com/Galabavamsi/sremu

OPEN-SOURCE CONTRIBUTIONS

HFlow, Hebbian Robotics (YC S26) github.com/Hebbian-Robotics/hflow Sep 2026 - Oct 2026
- 8 merged PRs (Python, Apache Arrow) to an SDK that verifies robot training data: LeRobot and robot-log conversion now keeps data whole or refuses it with a reason. Blocked silent 12- to 10-bit depth loss (PR #400), typed Arrow columns from every message (PR #657), fixed video-cache collisions (PR #353).

Inspect Robots, Robocurve (YC S26) github.com/robocurve/inspect-robots Sep 2026 - Oct 2026
- Evaluation framework for embodied-AI policies. 2 merged PRs: a configurable LLM-API retry policy across 5 wire protocols (PR #476) and unknown git state no longer logged as clean (PR #475); 3 open, including checkpoint/resume for long eval sets (PR #514).
- Wrote and maintain inspect-robots-wandb, a Weights & Biases plugin (pypi.org/project/inspect-robots-wandb): 100% coverage gate, CI on Python 3.10-3.13, trusted publishing.

Cerulion, Rust robot runtime 1 merged CLI fix (PR #216), 4 documentation fixes | github.com/cerulion-inc/cerulion Oct 2026

SKILLS

Languages: TypeScript/JavaScript, Python, SQL, Dart (Flutter), GLSL, Rust
Backend and frontend: Node.js (Fastify, Express), FastAPI, React, Flutter, Socket.IO, Redis, React Native (Expo), PostgreSQL (row-level security), Supabase, WebSockets, REST APIs
AI: LLM APIs, AI agents and tool use, MCP (Model Context Protocol), guardrails and evaluation, PyTorch
Infrastructure: Docker, GitHub Actions (OIDC), AWS Lightsail, Cloudflare Workers and D1, Render, Linux, Git, pytest
AI-assisted development: Claude Code; I set the design and verify results with tests and independent references.

ACHIEVEMENTS

- **Top 10 (Grand Jury round) of 4,500+ teams nationwide**, RAKSHAM AI Cybersecurity Hackathon (IIT Delhi, I4C and Amazon), Oct 2026; Pehchaan, team of 2.
- **5th of 232 teams**, WEAR Dataset Challenge @HASCA 2026 (Kaggle); **All India Rank 200** (2026) and **700** (2025), Amazon ML Challenge (team); **5th place**, Arista Networks Wi-Fi challenge, Inter IIT Tech Meet 14.0 (team); **1st place**, IIT Bhilai Web Development Competition.
- Prime Minister's Scholarship Scheme (PMSS) scholar, 2024; Subroto Mukherjee Scholarship (IAFBA), 2023; Top 1% in JEE Advanced; attended YC Startup School, Bengaluru.

PUBLICATIONS

- **Co-author**, *Inspect Robots: Evaluating the Capabilities and Safety of Embodied AI*, arXiv:2610.06306; SPAIS Workshop at CoRL 2026 (poster).
- **Co-author**, 2 RIS papers: IEEE INDICON 2025 and *Ad Hoc Networks* (Elsevier), 2026.